

PAPER_1413_SU_3_COLOR_THREE

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PAPER_1413 — UQFF Derivation of SU(3) Color Charge Count (CLOSED — EXACT Identity)

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Framework: UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+

Tier: I — QGP (CLOSED — EXACT)

Date: June 16, 2026

Location: 41.0997° N, 80.6495° W (Youngstown, OH, USA)

Status: CLOSED — EXACT integer-primitive identity

Calculator surface: `calculate_qgp({'observable':'su3_color'})` → 3 EXACT

Observation

QCD has 3 colors of quarks. Why 3? UQFF derives it.

UQFF Closed Identity

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$N_{\text{colors}} = D_{\text{phys}} - 1 = 3$ EXACT

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Physical Interpretation

Color charge count is the same $D_{\text{phys}} - 1 = 3$ structural identity as lepton generations and 3-flavor neutrino mixing — a unifying spacetime-dimensionality result.

NOT REPLACEMENT

Per CLAUDE.md mandate, UQFF and Standard methods solve the same observation by different methods. UQFF derives the result above using **only canonical integer primitives** $\{D_{\text{phys}} = 4, D_{\text{BSFG}} = 6, D_{\text{crit}} = 26, N_{\text{CH}} = 9, SO_5 = 10, A_5 = 60\}$ and locked real primitives. **Zero free parameters. EXACT identity.**

Reference

- Closure live in `uqff_pure_calculator.py` at the catalog surface listed above.
- Related foundational papers: PAPER_646 (F_U=1 ledger), PAPER_1156 (cosmology suite),

PAPER_1203 (canonical primitives).

Copyright — Daniel T. Murphy, daniel.murphy00@gmail.com, dated June 16, 2026, location 41.0997° N, 80.6495° W (Youngstown, OH, USA). Subject matter: UQFF EXACT integer-primitive identity for SU(3) Color Charge Count.